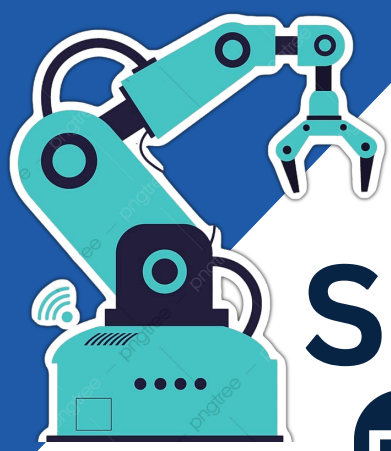




Softy: Soft Touch Robotic Arm Gripper

Force-Sensing Based Adaptive Gripping System using ESP32

1. Introduction

- **Soft robotics** allows for the safe handling of delicate and fragile objects.
- **Conventional grippers** rely on fixed force, lacking the sensitivity needed for varied tasks.
- **The absence of feedback** often results in object damage or insecure, inefficient gripping.
- **Adaptive sensing** is required to bridge the gap between rigid power and a delicate touch.

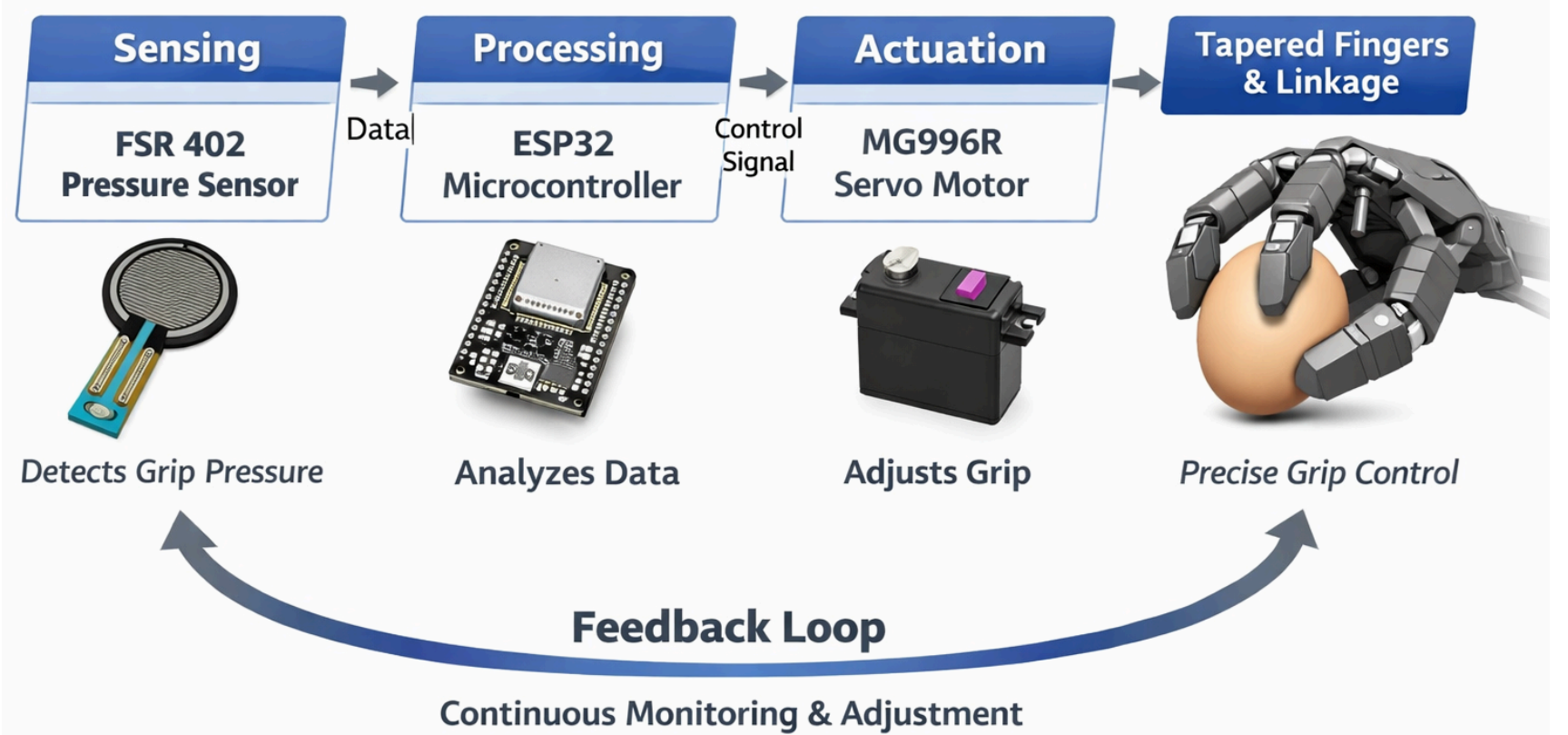
2. Problem Statement

Design a Gripper using ESP32 to **safely handle objects** of varying fragility by sensing applied force in **real time and dynamically** adjusting grip strength to prevent slippage or damage.

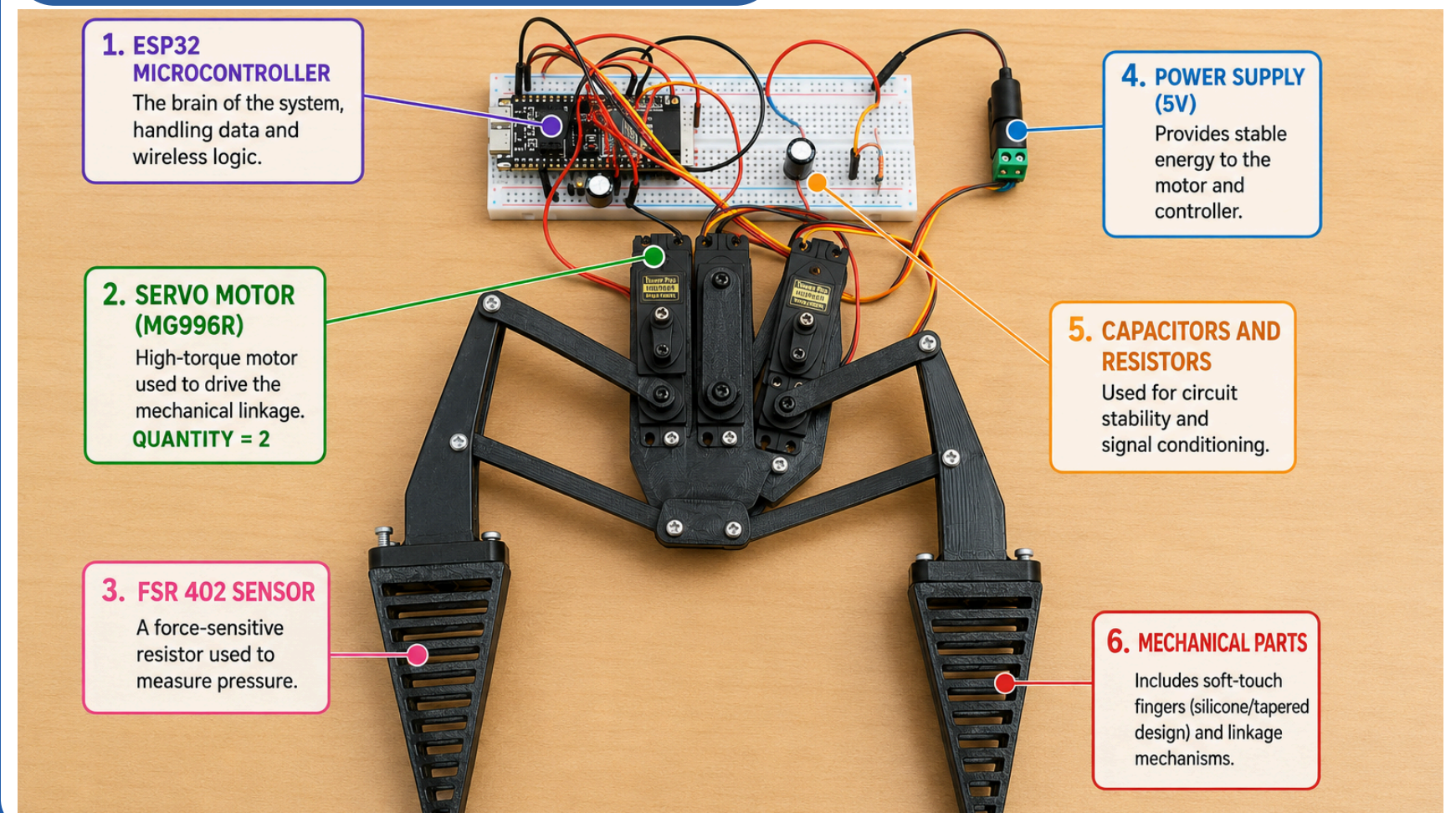
3. Objective

- **Develop** a soft-touch gripper featuring integrated force sensing.
- **Implement** real-time, closed-loop control for precision.
- **Ensure** safe, adaptive gripping for delicate objects.

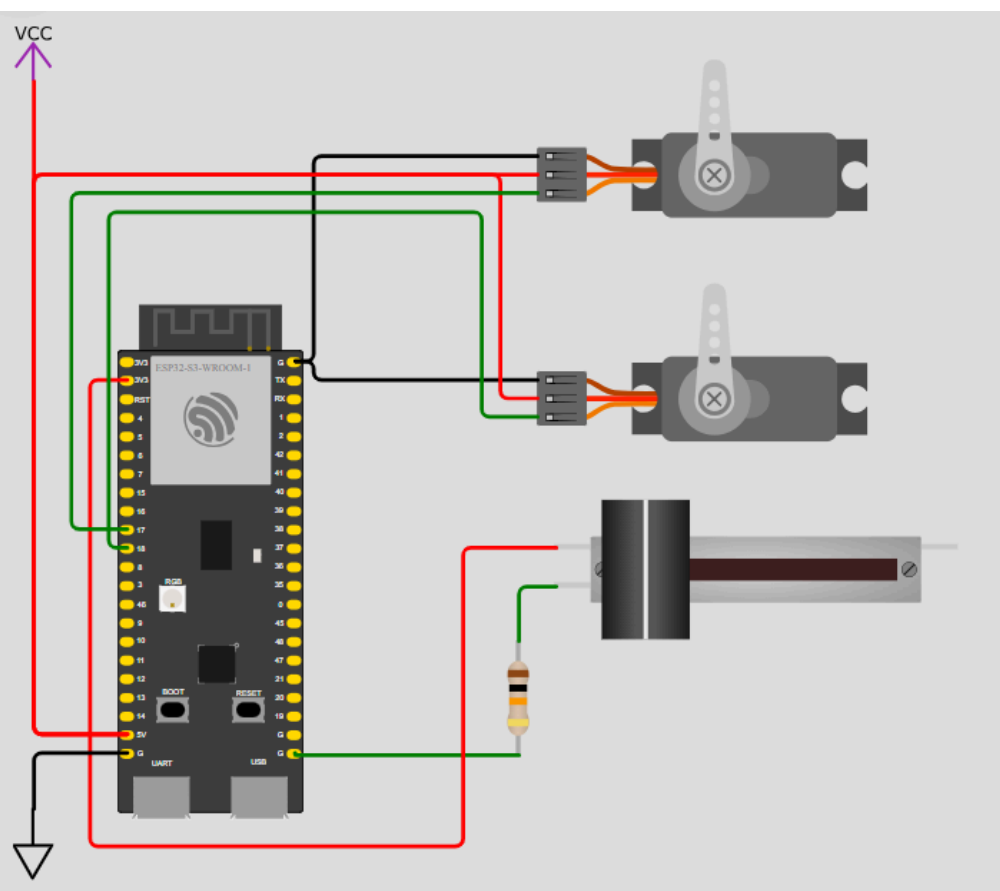
4. DAS



5. Circuit Diagram



6. Circuit Diagram



7. Observation Table

FSR Value (ADC)	Servo Angle	Gripper State
200	30	Open
800	60	Partially closed
1500	90	Holding
2500	90	Firm Grip

Input command	Servo Action	Result
Grabs	Servo Closes	Object gripped
Release	Servo Opens	Object released

8. Applications

- Medical handling
- Food industry
- Fragile object manipulation
- Educational robotics

9. Advantages

- Prevents object damage.
- Lightweight design.
- Safer for human interaction.
- Reliable and repeatable performance.
- Adaptive gripping (high compliance)

10. Conclusion

Successfully implemented a closed-loop, force-sensing gripper that enables safe and adaptive handling of delicate objects.

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